SAFETY DATA SHEET

DANAFIX 448

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Trade name

DANAFIX 448

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses of the substance or mixture

PVAc Glue

Relevant identified uses of the substance or mixture (REACH)

No special

Uses advised against

No special

1.3. Details of the supplier of the safety data sheet

Company and address

Dana Lim A/S
Københavnsvej 220
DK-4600 Køge
Denmark
Tel: +45 56 64 00 70
Fax: +45 56 64 00 90

Contact person

Product Safety Department

E-mail

info@danalim.dk

SDS date

2020-10-12

SDS Version

1.1

Date of previous version

2020-10-12 (1.0)

1.4. Emergency telephone number

Contact The National Poisons Information Service (dial 111, 24 h service).
See section 4 “First aid measures”.

SECTION 2: Hazards identification

2.1. Classification of the substance or mixture

-

2.2. Label elements

Hazard pictogram(s)

Not applicable

Signal word

Not applicable

Hazard statement(s)

Not applicable

Safety statement(s)

General

-

Prevention

-

Response

-

Storage

-

Disposal

-

Hazardous substances

No special

2.3. Other hazards

Additional labelling

EUH208, Contains (S)-p-mentha-1,8-diene;trans-1-methyl-4-(1-methylvinyl)cyclohexene;(R)-p-mentha-1,8-diene;dipentene;(±)-1-methyl-4-(1-methylvinyl)cyclohexene;d-limonene;limonene;l-limonene, Reaction mass of: 5-chloro-2-methyl-4-isothiazolin-3-one and 2-methyl-2H-isothiazol-3-one (3:1), 2,4,7,9-tetramethyldec-5-yne-4,7-diol. May produce an allergic reaction.

Active substance(s):

Reaction mass of: 5-chloro-2-methyl-4-isothiazolin-3-one and 2-methyl-2H-isothiazol-3-one (3:1) (0.0014 g/100g)

Additional warnings

This mixture/product does not contain any substances considered to meet the criteria classifying them as PBT and/or vPvB.

SECTION 3: Composition/information on ingredients

3.2 Mixtures

Product/Ingredient name	Identifiers	% w/w	Classification	Note
propylene carbonate	CAS No.: 108-32-7 EC No.: 203-572-1 REACH No.: 01-2119537232-48-XXXX Index No.: 607-194-00-1	3-5%	Eye Irrit. 2, H319	
(S)-p-mentha-1,8-diene;trans-1-methyl-4-(1-methylvinyl)cyclohexene;(R)-p-mentha-1,8-diene;dipentene;(±)-1-methyl-4-(1-methylvinyl)cyclohexene;d-limonene;limonene;l-limonene	CAS No.: 5989-27-5 EC No.: 227-813-5 REACH No.: 01-2119529223-47-XXXX Index No.: 601-029-00-7	<1%	Flam. Liq. 3, H226 Skin Irrit. 2, H315 Skin Sens. 1, H317 Aquatic Chronic 1, H410 (M=1) Asp. Tox. 1, H304 Aquatic Acute 1, H400 (M=1)	
2,4,7,9-tetramethyldec-5-yne-4,7-diol	CAS No.: 126-86-3 EC No.: 204-809-1 REACH No.: Index No.:	<1%	Skin Sens. 1B, H317 Eye Dam. 1, H318 Aquatic Chronic 3, H412	
Reaction mass of: 5-chloro-2-methyl-4-isothiazolin-3-one and 2-methyl-2H-isothiazol-3-one (3:1)	CAS No.: 55965-84-9 EC No.: 911-418-6 REACH No.: 01-	<0.0015%	Acute Tox. 3, H301 Acute Tox. 2, H310 Skin Corr. 1C, H314 (SCL: 0.60 %) Skin Sens. 1A, H317 (SCL: 0.0015 %)	

	2120764691-48-XXXX Index No.:	Eye Dam. 1, H318 Acute Tox. 2, H330 Aquatic Acute 1, H400 (M=100) Aquatic Chronic 1, H410 (M=100) EUH071

See full text of H-phrases in section 16. Occupational exposure limits are listed in section 8, if these are available.

Other information

EU: European occupational exposure limit

SECTION 4: First aid measures

4.1. Description of first aid measures

General information

In the case of accident: Contact a doctor or casualty department – take the label or this safety data sheet. Contact a doctor if in doubt about the injured person's condition or if the symptoms persist. Never give an unconscious person water or other drink.

Inhalation

Upon breathing difficulties or irritation of the respiratory tract: Bring the person into fresh air and stay with him/her.

Skin contact

Immediately remove contaminated clothing and shoes. Ensure that skin, which has been exposed to the material, is washed thoroughly with water and soap. Skin cleanser can be used. DO NOT use solvents or thinners.

Eye contact

Upon irritation of the eye: Remove contact lenses and open eyes widely. Flush eyes with water or saline water(20-30°C) for at least 5 minutes. Seek medical assistance and continue flushing during transport.

Ingestion

Provide plenty of water for the person to drink and stay with him/her. In case of malaise, seek medical advice immediately and bring the safety data sheet or label from the product. Do not induce vomiting, unless recommended by the doctor. Have the victim lean forward with head down to avoid inhalation of- or choking on vomited material.

Burns

Not applicable

4.2. Most important symptoms and effects, both acute and delayed

This product contains substances that may trigger an allergic reaction to predisposed persons.

4.3. Indication of any immediate medical attention and special treatment needed

No special

Information to medics

Bring this safety data sheet or the label from this product.

SECTION 5: Firefighting measures

5.1. Extinguishing media

Not applicable

5.2. Special hazards arising from the substance or mixture

Fire will result in dense smoke. Exposure to combustion products may harm your health. Closed containers, which are exposed to fire, should be cooled with water. Do not allow fire-extinguishing water to enter the sewage system and nearby surface waters.

5.3. Advice for firefighters

Fire fighters should wear appropriate personal protective equipment.

SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

No specific requirements

6.2. Environmental precautions

Avoid discharge to lakes, streams, sewers, etc.

6.3. Methods and material for containment and cleaning up

Use sand, sawdust, earth, vermiculite, diatomaceous earth to contain and collect non-combustible absorbent materials and place in container for disposal, according to local regulations.
To the extent possible cleaning is performed with normal cleaning agents. Avoid use of solvents.

6.4. Reference to other sections

See section on “Disposal considerations” in regard of handling of waste.
See section on 'Exposure controls/personal protection' for protective measures.

SECTION 7: Handling and storage

7.1. Precautions for safe handling

Smoking, drinking and consumption of food is not allowed in the work area.
See section on 'Exposure controls/personal protection' for information on personal protection.

7.2. Conditions for safe storage, including any incompatibilities

Always store in containers of the same material as the original container.
Containers that have been opened must be carefully resealed and kept upright to prevent leakage.
Must be stored in a cool and well-ventilated area, away from possible sources of ignition.

Storage temperature

Temperature
> 0°C

Incompatible materials

Strong acids, strong bases, strong oxidizing agents, and strong reducing agents.

7.3. Specific end use(s)

This product should only be used for applications quoted in section 1.2

SECTION 8: Exposure controls/personal protection

8.1. Control parameters

—
sulphur dioxide
Long term exposure limit (8 hours): 0,5 ppm
Long term exposure limit (8 hours): 1,3 mg/m³
Short term exposure limit (15 minutes): 1 ppm
Short term exposure limit (15 minutes): 2,7 mg/m³

The Control of Substances Hazardous to Health Regulations 2002. SI 2002/2677 The Stationery Office 2002. EH40/2005 Workplace exposure limits (Fourth Edition 2020)

DNEL

Product/Ingredient name	DNEL	Route of exposure	Duration
2,4,7,9-tetramethyldec-5-yne-4,7-diol	1,76 mg/m3	Inhalation	Long term – Systemic effects - Workers
2,4,7,9-tetramethyldec-5-yne-4,7-diol	5,28 mg/m3	Inhalation	Short term – Systemic effects - Workers

2,4,7,9-tetramethyldec-5-yne-4,7-diol	0,5 mg/kg	Dermal	Long term – Systemic effects - Workers
2,4,7,9-tetramethyldec-5-yne-4,7-diol	1,5 mg/kg	Dermal	Short term – Systemic effects - Workers
2,4,7,9-tetramethyldec-5-yne-4,7-diol	0,43 mg/m3	Inhalation	Long term – Systemic effects - General population
2,4,7,9-tetramethyldec-5-yne-4,7-diol	1,29 mg/m3	Inhalation	Short term – Systemic effects - General population
2,4,7,9-tetramethyldec-5-yne-4,7-diol	0,25 mg/kg	Dermal	Long term – Systemic effects - General population
2,4,7,9-tetramethyldec-5-yne-4,7-diol	0,75 mg/kg	Dermal	Short term – Systemic effects - General population
2,4,7,9-tetramethyldec-5-yne-4,7-diol	0,25 mg/kg	Oral	Long term – Systemic effects - General population
2,4,7,9-tetramethyldec-5-yne-4,7-diol	0,75 mg/kg	Oral	Short term – Systemic effects - General population

PNEC

Product/Ingredient name	PNEC	Route of exposure	Duration of Exposure
2,4,7,9-tetramethyldec-5-yne-4,7-diol	0,04 mg/l	Freshwater	No data available
2,4,7,9-tetramethyldec-5-yne-4,7-diol	0,004 mg/l	Marine water	No data available
2,4,7,9-tetramethyldec-5-yne-4,7-diol	0,4 mg/l	Intermittent release	No data available
2,4,7,9-tetramethyldec-5-yne-4,7-diol	7 mg/l	Sewage Treatment Plant	No data available
2,4,7,9-tetramethyldec-5-yne-4,7-diol	0,32 mg/kg dry weight	Freshwater sediment	No data available
2,4,7,9-tetramethyldec-5-yne-4,7-diol	0,032 mg/kg dry weight	Marine water sediment	No data available
2,4,7,9-tetramethyldec-5-yne-4,7-diol	0,028 mg/kg dry weight	Soil	No data available

8.2. Exposure controls

Compliance with the given occupational exposure limits values should be controlled on a regular basis.

▼General recommendations

Small amounts of SO2 may be formed if stored over a longer period of time. If possible use local exhaust
Smoking, eating and drinking are not allowed in the work premises

Exposure scenarios

There are no exposure scenarios implemented for this product.

Exposure limits

Professional users are subjected to the legally set maximum concentrations for occupational exposure. See occupational hygiene limit values above.

Appropriate technical measures

Airborne gas and dust concentrations must be kept at a minimum and below current limit values (see above). Installation of a local exhaust system if normal air flow in the work room is not sufficient is recommended. Ensure emergency eyewash and -showers are clearly marked.

Hygiene measures

In between use of the product and at the end of the working day all exposed areas of the body must be washed thoroughly. Always wash hands, forearms and face.

Measures to avoid environmental exposure

No specific requirements

Individual protection measures, such as personal protective equipment

Generally

Use only CE marked protective equipment.

Respiratory Equipment

No specific requirements

Skin protection

No specific requirements

Hand protection

Work situation	Material	Glove thickness (mm)	Breakthrough time (min.)	Standards	
	Nitrile	0.1	> 480	EN374-2, EN388	

Eye protection

No specific requirements

SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Form

Liquid

Colour

White

Odour

Characteristic

Odour threshold (ppm)

Testing not relevant or not possible due to nature of the product.

pH

3

Density (g/cm³)

1.10

Viscosity

4800-6000 mPa.s

Phase changes

Melting point (°C)

Testing not relevant or not possible due to nature of the product.

Boiling point (°C)

100.00 °C

Vapour pressure

Testing not relevant or not possible due to nature of the product.

Vapour density

Testing not relevant or not possible due to nature of the product.

Decomposition temperature (°C)

Testing not relevant or not possible due to nature of the product.

Evaporation rate (n-butylacetate = 100)

Testing not relevant or not possible due to nature of the product.

Data on fire and explosion hazards

Flash point (°C)

Testing not relevant or not possible due to nature of the product.

Ignition (°C)

Testing not relevant or not possible due to nature of the product.

Auto flammability (°C)

Testing not relevant or not possible due to nature of the product.

Explosion limits (% v/v)

Testing not relevant or not possible due to nature of the product.

Explosive properties

Testing not relevant or not possible due to nature of the product.

Oxidizing properties

Testing not relevant or not possible due to nature of the product.

Solubility

Solubility in water

Soluble

n-octanol/water coefficient

Testing not relevant or not possible due to nature of the product.

Solubility in fat (g/L)

Testing not relevant or not possible due to nature of the product.

9.2. Other information

SECTION 10: Stability and reactivity

10.1. Reactivity

No data available

10.2. Chemical stability

The product is stable under the conditions, noted in the section “Handling and storage”.

10.3. Possibility of hazardous reactions

No special

10.4. Conditions to avoid

No special

10.5. Incompatible materials

Strong acids, strong bases, strong oxidizing agents, and strong reducing agents.

10.6. Hazardous decomposition products

The product is not degraded when used as specified in section 1.

SECTION 11: Toxicological information

11.1. Information on toxicological effects

Acute toxicity

Product/Ingredient name	Species	Test	Route of exposure	Result
Reaction mass of: 5-chloro-2-methyl-4-isothiazolin-3-one and 2-methyl-2H-isothiazol-3-one (3:1)	Rat	LD50	Oral	49,6-75 mg/kg ·
Reaction mass of: 5-chloro-2-methyl-4-isothiazolin-3-one and 2-	Rat	LC50	Inhalation	0,33 mg/l, 4 h aerosol ·

methyl-2H-isothiazol-3-one (3:1)				
Reaction mass of: 5-chloro-2-methyl-4-isothiazolin-3-one and 2-methyl-2H-isothiazol-3-one (3:1)	Rabbit	LD50	Dermal	141 mg/kg ·

Skin corrosion/irritation
Based on available data, the classification criteria are not met.
Serious eye damage/irritation
Based on available data, the classification criteria are not met.
Respiratory or skin sensitisation
Based on available data, the classification criteria are not met.
This product contains substances that may trigger an allergic reaction to predisposed persons.
Germ cell mutagenicity
Based on available data, the classification criteria are not met.
Carcinogenicity
Based on available data, the classification criteria are not met.
Reproductive toxicity
Based on available data, the classification criteria are not met.
STOT-single exposure
Based on available data, the classification criteria are not met.
STOT-repeated exposure
Based on available data, the classification criteria are not met.
Aspiration hazard
Based on available data, the classification criteria are not met.
Long term effects
No special
Other information
sulphur dioxide has been classified by IARC as a group 3 carcinogen.

SECTION 12: Ecological information

12.1. Toxicity

Product/Ingredient name	Species	Test	Duration	Result
Reaction mass of: 5-chloro-2-methyl-4-isothiazolin-3-one and 2-methyl-2H-isothiazol-3-one (3:1)	Algae	EC50	72 hours	0,027 mg/l ·

12.2. Persistence and degradability

Product/Ingredient name	Biodegradability	Test	Result
Reaction mass of: 5-chloro-2-methyl-4-isothiazolin-3-one and 2-methyl-2H-isothiazol-3-one (3:1)	Yes	OECD 301 D	>60%

12.3. Bioaccumulative potential

Product/Ingredient name	Potential bioaccumulation	LogPow	BCF

Reaction mass of: 5-chloro-2-methyl-4-isothiazolin-3-one and 2-methyl-2H-isothiazol-3-one (3:1)	No	No data available	3.6000000

12.4. Mobility in soil
No data available

12.5. Results of PBT and vPvB assessment
This mixture/product does not contain any substances considered to meet the criteria classifying them as PBT and/or vPvB.

12.6. Other adverse effects
This product contains substances that are toxic to the environment. May result in adverse effects to aquatic organisms.

This product contains substances, which may cause adverse long-term effects to the aquatic environment.

SECTION 13: Disposal considerations

13.1. Waste treatment methods
Product is not covered by regulations on dangerous waste.

EWC code

08 04 10	Waste adhesives and sealants other than those mentioned in 08 04 09

Specific labelling
Not applicable

Contaminated packing
Packaging containing residues of the product must be disposed of similarly to the product.

SECTION 14: Transport information

14.1 - 14.4
Not dangerous goods according to ADR, IATA and IMDG.

ADR/RID
Not applicable

IMDG
Not applicable

IATA
Not applicable

"MARINE POLLUTANT"
No

14.5. Environmental hazards
Not applicable

14.6. Special precautions for user
Not applicable

14.7. Transport in bulk according to Annex II of Marpol and the IBC Code
No data available

SECTION 15: Regulatory information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Restrictions for application
Restricted to professional users.

Demands for specific education
No specific requirements

SEVESO - Categories / dangerous substances

Not applicable

Additional information

Not applicable

Sources

Regulation (EU) No 528/2012 of the European Parliament and of the Council of 22 May 2012 concerning the making available on the market and use of biocidal products.
Regulation (EC) No 1272/2008 of the European Parliament and of the Council of 16 December 2008 on classification, labelling and packaging of substances and mixtures, amending and repealing Directives 67/548/EEC and 1999/45/EC, and amending Regulation (EC) No 1907/2006 (CLP).
Regulation (EC) 1907/2006 (REACH).

15.2. Chemical safety assessment

No

SECTION 16: Other information

Full text of H-phrases as mentioned in section 3

H319, Causes serious eye irritation.
H226, Flammable liquid and vapour.
H315, Causes skin irritation.
H317, May cause an allergic skin reaction.
H410, Very toxic to aquatic life with long lasting effects.
H304, May be fatal if swallowed and enters airways.
H400, Very toxic to aquatic life.
H318, Causes serious eye damage.
H412, Harmful to aquatic life with long lasting effects.
H301, Toxic if swallowed.
H310, Fatal in contact with skin.
H314, Causes severe skin burns and eye damage.
H330, Fatal if inhaled.
EUH071, Corrosive to the respiratory tract.

Abbreviations and acronyms

ADN = European Provisions concerning the International Carriage of Dangerous Goods by Inland Waterway
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road
ATE = Acute Toxicity Estimate
BCF = Bioconcentration Factor
CAS = Chemical Abstracts Service
CLP = Classification, Labelling and Packaging Regulation [Regulation (EC) No. 1272/2008]
CSA = Chemical Safety Assessment
CSR = Chemical Safety Report
DMEL = Derived Minimal Effect Level
DNEL = Derived No Effect Level
EINECS = European Inventory of Existing Commercial chemical Substances
ES = Exposure Scenario
EUH statement = CLP-specific Hazard statement
EWC = European Waste Catalogue
GHS = Globally Harmonized System of Classification and Labelling of Chemicals
IARC = International Agency for Research on Cancer (IARC)
IATA = International Air Transport Association
IBC = Intermediate Bulk Container
IMDG = International Maritime Dangerous Goods
LogPow = logarithm of the octanol/water partition coefficient
MARPOL = International Convention for the Prevention of Pollution From Ships, 1973 as modified by the Protocol of 1978. ("Marpol" = marine pollution)
OECD = Organisation for Economic Co-operation and Development
PBT = Persistent, Bioaccumulative and Toxic

PNEC = Predicted No Effect Concentration
RID = The Regulations concerning the International Carriage of Dangerous Goods by Rail
RRN = REACH Registration Number
SCL = A specific concentration limit.
SVHC = Substances of Very High Concern
STOT-RE = Specific Target Organ Toxicity - Repeated Exposure
STOT-SE = Specific Target Organ Toxicity - Single Exposure
TWA = Time weighted average
UN = United Nations
UVCB = Complex hydrocarbon substance
VOC = Volatile Organic Compound
vPvB = Very Persistent and Very Bioaccumulative

Additional information

Not applicable

The safety data sheet is validated by

Product Safety Department

Other

A change (in proportion to the last essential change (first cipher in SDS version, see section 1)) is marked with a blue triangle.
The information in this safety data sheet applies only to this specific product (mentioned in section 1) and is not necessarily correct for use with other chemicals/products.
It is recommended to hand over this safety data sheet to the actual user of the product. Information in this safety data sheet cannot be used as a product specification.